

Dungeon Escape — test case document

Test cases covering the core game logic in `dungeon_escape.py`: startup, combat, healing, running, looting, weapon upgrades, win/loss conditions, and the Play Again reset. Test 13 is highlighted as a known bug.

#	Test case	Steps / input	Expected result
1	Start with valid name	Enter “Alex”, click Enter the dungeon	Game screen shows, player has 60 HP, inventory = [Health Potion, Wooden Sword], Room 1 (Slime) begins
2	Start with blank name	Leave name field empty, click Enter the dungeon	Defaults to “Eliezer EJ A. Maique” instead of blank
3	Attack with a weapon	Click Attack while Wooden Sword is in inventory	Damage roll is $10-15 + (\text{weapon_level} - 1)$; enemy HP drops; if enemy survives, player takes enemy's fixed damage
4	Attack without a weapon	Remove/never have a sword, click Attack	Damage roll falls back to 6–12
5	Critical hit	Click Attack repeatedly (~15% chance per hit)	Log shows “Critical strike!” and damage is doubled that turn
6	Heal with potion available	Click Heal with a Health Potion in inventory	Potion is consumed, HP +20 (capped at 60), score +5
7	Heal with no potion	Use up all potions, click Heal again	Log shows “No potions left!”, no HP/score change, no potion consumed
8	Defeat an enemy	Reduce enemy HP to 0 via Attack	Loot added to inventory, score +50 + loot bonus, log shows loot message, next room begins
9	Player HP reaches 0	Let enemy damage bring health to ≤ 0	<code>finish_game</code> (“The dungeon defeated you. GAME OVER.”) fires, action buttons disable
10	Run from a room	Click Run during any battle	Log shows escape message, <code>enemy_index</code> increments, next room begins (no loot, no damage)
11	Defeat all 4 enemies	Win every battle through Shadow Dragon	<code>finish_game</code> (“...escaped the dungeon!”) fires, victory popup shown
12	Weapon upgrade at room 3	Defeat the Skeleton King (<code>enemy_index</code> becomes 2) at <code>weapon_level</code> 1	Iron Sword added, <code>weapon_level</code> set to 2, log shows upgrade message
13	Weapon upgrade at room 5 (BUG)	Defeat all enemies, reach <code>enemy_index == 4</code>	Never triggers — <code>ENEMIES</code> only has indices 0–3, so the Dragon Blade check is unreachable
14	Loot drop randomness	Defeat several enemies	Loot is one of Health Potion / Gold Coin / Magic Scroll; a Gold Coin also adds 10 to <code>self.gold</code>
15	Play Again resets state	Finish a game (win or lose), click Play Again	Health, inventory, score, gold, weapon level, and enemy index all reset; start screen reappears
16	<code>find_item</code> with missing item	Attack with no sword variants in inventory	Returns <code>None</code> , damage falls back to the unarmed 6–12 range

Note: Test 13 exposes a real bug — the Dragon Blade upgrade condition checks `enemy_index == 4`, but `ENEMIES` only has 4 entries (indices 0–3), so the game ends via victory before that index is ever reached.